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Q1

∵ Irreducible & Recurrent : $\exists$ infinite n, $P_{ii}^{n}=1$; $\exists m$, $P_{ij}^{m}>0$

∴ In the long run, every t times, Markov chain reenter i, there is a positive probability to enter j.

Assume after s times of visiting i, still haven't visit j:

$P_{(s)}=(1-P_{ij}^{m})^{s}$ ∵ $P_{ij}^{m}>0$ ∴ $\lim_{s\to\infty}(1-P_{ij}^{m})^{s}=0$

∴ There must be at least once to enter j, $f_{ij}=1$

Q2

(1) Starting at i
- Arrival (λ)
  - Demand ($\lambda\mu$)
  - No demand ($\lambda(1-\mu)$)
- No arrival ($1-\lambda$)
  - Demand
    - $i>0$: $(1-\lambda)\mu$
    - $i=0$: ↓ contributes to staying at 0
  - No demand ($(1-\lambda)(1-\mu)$)

For $i=0$
- To 0
  - Arrival & Demand : $\lambda\mu$
  - No arrival & Demand : $(1-\lambda)\mu$
  - No arrival & No demand : $(1-\lambda)(1-\mu)$
- To 1 : Arrival & No demand : $\lambda(1-\mu)$
- To $j\geq 2$: 0

$$\Longrightarrow P=\begin{bmatrix} 1-\lambda(1-\mu) & \lambda(1-\mu) & 0 & 0 & \cdots \\ (1-\lambda)\mu & \lambda\mu+(1-\lambda)(1-\mu) & \lambda(1-\mu) & 0 & \cdots \\ 0 & (1-\lambda)\mu & \lambda\mu+(1-\lambda)(1-\mu) & \lambda(1-\mu) & \cdots \\ 0 & 0 & (1-\lambda)\mu & \lambda\mu+(1-\lambda)(1-\mu) & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Irreducibility : $\forall$ state i,j (WLOG, assume $j>i$), $P_{ij}^{j-i}=[\lambda(1-\mu)]^{j-i}>0$

∴ Irreducible

(2) $P(i,j)=\begin{cases} \frac{\lambda(1-\mu)}{\lambda(1-\mu)+(1-\lambda)\mu} & ; \text{if } j=i+1 \\ \frac{(1-\lambda)\mu}{\lambda(1-\mu)+(1-\lambda)\mu} & , \text{if } j=i-1\geq 0 \\ & , \text{if } i=0 \ \& \ i=1 \end{cases}$

$$P_Y = \begin{bmatrix} 0 & 1 & 0 & 0 & \cdots \\ \frac{(1-\lambda)\mu}{\lambda(1-\mu)+(1-\lambda)\mu} & 0 & \frac{\lambda(1-\mu)}{\lambda(1-\mu)+(1-\lambda)\mu} & 0 & \cdots \\ 0 & \frac{(1-\lambda)\mu}{\lambda(1-\mu)+(1-\lambda)\mu} & 0 & \frac{\lambda(1-\mu)}{\lambda(1-\mu)+(1-\lambda)\mu} & \cdots \\ 0 & 0 & \frac{(1-\lambda)\mu}{\lambda(1-\mu)+(1-\lambda)\mu} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

(3) Let $p = \frac{\lambda(1-\mu)}{\lambda(1-\mu)+(1-\lambda)\mu}$, $q_0 = \frac{(1-\lambda)\mu}{\lambda(1-\mu)+(1-\lambda)\mu} = 1-p$

- $p=q_0$: $\lambda=\mu$, $p=\frac{1}{2}$, symmetric random walk, null recurrent
- $p>q_0$: $\lambda>\mu$, $p>\frac{1}{2}$, drift up, transient
- $p<q_0$: $\lambda<\mu$, $p<\frac{1}{2}$, drife down, positive recurrent

(4)
- $p=q_0$: $\lambda=\mu$, null recurrent
- $p>q_0$: $\lambda>\mu$, transient
- $p<q_0$: $\lambda<\mu$, positive recurrent

(5) $\begin{cases} \pi = \pi P \\ \pi \mathbb{1} = 1 \end{cases} \Rightarrow \begin{cases} \pi_0 = \pi_0[1-(1-\mu)\lambda] + \pi_1[(1-\lambda)\mu] \\ \pi_1 = \pi_0[\lambda(1-\mu)] + \pi_1[\lambda(1-\mu)+(1-\lambda)\mu] + \pi_2[(1-\lambda)\mu] \end{cases}$

$\Rightarrow \frac{\pi_1}{\pi_0} = \frac{\lambda(1-\mu)}{\mu(1-\lambda)} = \frac{\pi_2}{\pi_1}$ Let $\frac{\pi_{i+1}}{\pi_i} = \frac{\lambda(1-\mu)}{(1-\lambda)\mu} = t$

$\pi_0(1+t+t^2+\cdots) = \frac{\pi_0}{1-t} = 1 \quad \therefore \pi_0 = 1-t$

$\therefore \Pi = \left(1-t, (1-t)t, (1-t)t^2, \cdots, (1-t)t^n, \cdots\right)$

Q3

(1) $P_i = q_{0i} P P_{i+1} + q_{0i}(1-p) P_{i-1} + (1-q_{0i}) P_i$

$\therefore P_i = p P_{i+1} + (1-p) P_{i-1}$, which is the form of gambling ruin model.

$P_i = \begin{cases} \frac{1-(\frac{1}{p}-1)^i}{1-(\frac{1}{p}-1)^N}, & p \neq \frac{1}{2} \\ i/N, & p=\frac{1}{2} \end{cases}$ $P_i' = \begin{cases} \frac{1-(\frac{p}{1-p})^i}{1-(\frac{p}{1-p})^{n+1-N}}, & p \neq \frac{1}{2} \\ i/n+1-N, & p=\frac{1}{2} \end{cases}$

$P = q_0 P P_1 + q_0(1-p) P_1' = \begin{cases} q_0, \text{ if } p=0 \text{ or } 1, \text{ because it move along one direction} \\ q_0 p \frac{1-(\frac{1}{p}-1)}{1-(\frac{1}{p}-1)^i} + q_0(1-p)\frac{1-\frac{p}{1-p}}{1-(\frac{p}{1-p})^{n+1-i}}, & \text{if } p \neq \frac{1}{2} \\ q_0 p \frac{1}{2} + q_0(1-p)\frac{1}{n+1-i}, & \text{if } p \in (0,1)\setminus\{\frac{1}{2}\} \end{cases}$

(2) If move to 1 first: set $i=n$

$$P_1 = \begin{cases} g_0 P \dfrac{1-(\frac{1}{p}-1)}{1-(\frac{1-p}{p})^n}, & \text{if } p\in(0,1) \text{ \& } p\neq\frac{1}{2} \\ g_0 P \frac{1}{n}, & \text{if } p=\frac{1}{2} \\ g_0 P, & \text{if } p=0 \text{ or } 1 \end{cases}$$

If move to 2 first: set $i=1$

$$P_2 = \begin{cases} g_0(1-p)\dfrac{1-\frac{p}{1-p}}{1-(\frac{p}{1-p})^n}, & \text{if } p\in(0,1) \text{ \& } p\neq\frac{1}{2} \\ g_0(1-p)\frac{1}{n}, & \text{if } p=\frac{1}{2} \\ g_0(1-p), & \text{if } p=0 \text{ or } 1 \end{cases}$$

$$\therefore P = \begin{cases} g_0 P \dfrac{1-(\frac{1}{p}-1)}{1-(\frac{1}{p}-1)^n} + g_0(1-p)\dfrac{1-(\frac{p}{1-p})}{1-(\frac{p}{1-p})^n}, & \text{if } p\in(0,1) \text{ \& } p\neq\frac{1}{2} \\ g_0\frac{1}{n}, & \text{if } p=\frac{1}{2} \\ g_0, & \text{if } p=0 \text{ or } 1 \end{cases}$$

(3) $p=\frac{1}{2}: \frac{1}{2}$ $p=0:0$ $p=1:1$

$p\in(0,1)$: $P($move to i first $|$ come back before inspecting all$)$

(A: move to i first; B: come back before inspecting all)

$$= \frac{P(A,B)}{P(B)} = \frac{\frac{2p-1}{1-(\frac{1-p}{p})^n}}{\frac{2p-1}{1-(\frac{1-p}{p})^n} + \frac{1-2p}{1-(\frac{p}{1-p})^n}} = \frac{1}{1+(\frac{1-p}{p})^n}$$

$$\therefore P' = \begin{cases} \frac{1}{2}, & p=\frac{1}{2} \\ 0, & p=0 \\ 1, & p=1 \\ \dfrac{1}{1+(\frac{1-p}{p})^n}, & p\in(0,1) \text{ \& } p\neq\frac{1}{2} \end{cases}$$

Q4

$$E\left[\sum_{k=0}^{\infty} X_k \mid X_0=n\right] = \sum_{k=0}^{\infty} E[X_k \mid X_0=n] = \sum_{k=0}^{\infty} n\mu^k = \frac{n}{1-\mu}$$